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climepi: A Python package for modeling climate suitability for vector-borne diseases

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Software

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Summary

climepi (clim-epi) is a Python package for combining climate projections with models of climate suitability for vector-borne diseases (VBDs). Utilizing the xarray library (Hoyer & Hamman, 2017) for handling labeled multi-dimensional arrays, climepi provides methods for: accessing climate projection data from a range of sources; defining, parameterizing, and running models of climate suitability for VBDs; and assessing the impacts of different sources of climate uncertainty (uncertainty in emissions scenarios, structural uncertainty across climate models, and natural climate fluctuations that occur alongside anthropogenic climate change). The package also provides a front-end application that can be used to explore impacts of climate change and its uncertainties using example climate datasets and climate-VBD suitability models (a web interface is available).

Statement of need

Climate change is altering the dynamics of a range of infectious diseases, particularly VBDs such as malaria and dengue (Ryan et al., 2019). Quantitative estimates of future VBD risks are important for targeting vector and pathogen surveillance resources, and for planning public health interventions. However, a recent review paper highlighted that few operational software tools exist for assessing climate-sensitive disease risks (Ryan et al., 2023). climepi contributes to addressing this gap by providing a flexible and extensible Python package, as well as a user-friendly front-end application, which can be used by climate-health researchers and other users (such as public health professionals) to assess future climate suitability for VBDs and uncertainty therein.

Main features

climepi is designed with a modular structure, comprising four components:

1. Climate data (`climdata`) subpackage: enables users to access climate projection data from different sources through a single interface. Rather than providing comprehensive access options for a large number of datasets, the focus is on facilitating straightforward access to data sources and climate variables (temperature and precipitation) that are particularly useful for analyzing the impacts of different types of climate uncertainty on future climate-VBD suitability. Supported data sources include the Inter-Sectoral Impact Model Intercomparison Project (ISIMIP) (Lange & Büchner, 2021), which provides downscaled and bias-adjusted data from multiple Coupled Model Intercomparison Project Phase 6 (CMIP6) (Eyring et al., 2016) climate models and emissions scenarios, and the Community Earth System Model version 2 Large Ensemble (LENS2) (Rodgers et al., 2021), which provides 100 ensemble members for analyzing internal (natural) climate variability.

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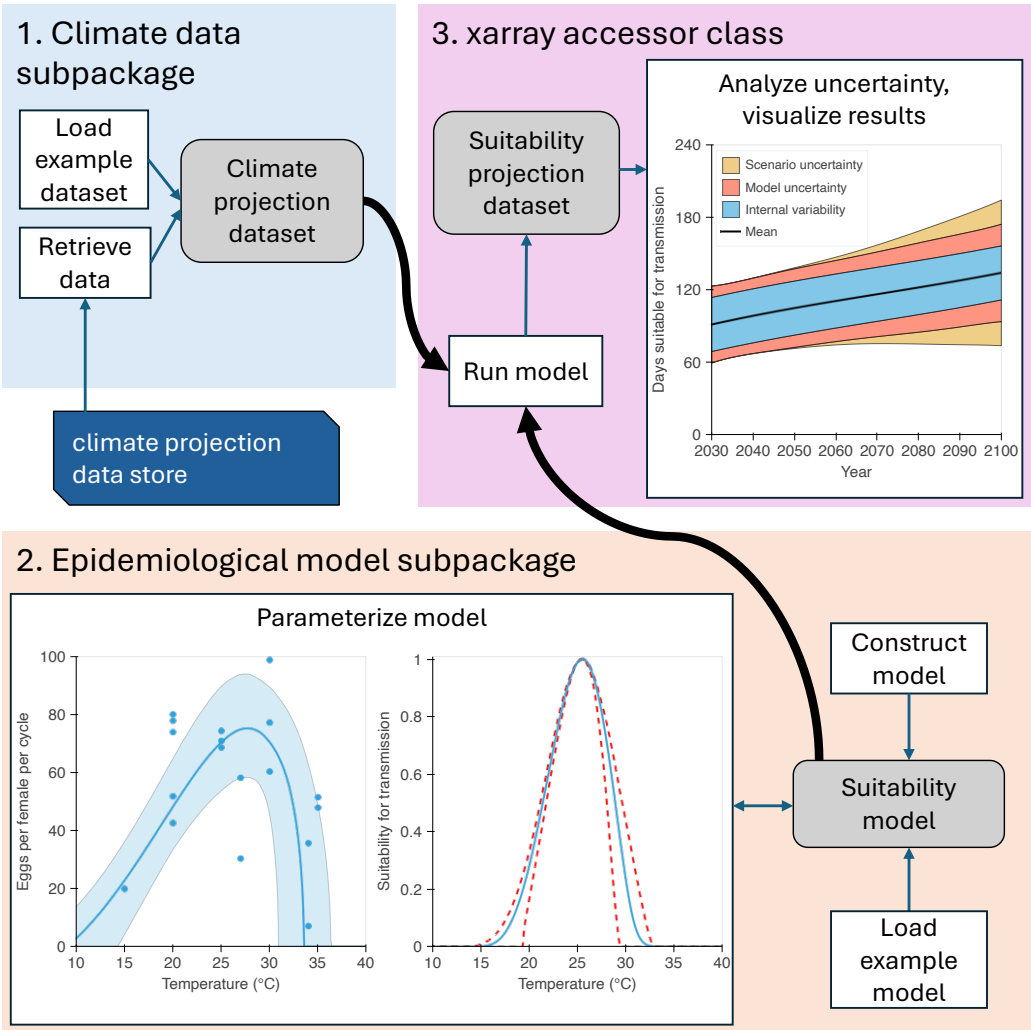

Figure 1: Schematic illustrating typical workflows using climepi. (1) The `climdata` subpackage can be used to access climate data from an external source, or to load a pre-defined example climate dataset. (2) The `epimod` subpackage can be used to define a mechanistic model describing climate suitability for VBD, infer temperature response curves of trait parameters using laboratory data (left plot), and determine the overall temperature dependence of the suitability metric (right plot). Pre-defined epidemiological models from the literature are also available as built-in examples. (3) The `climepi` accessor for xarray datasets can be used to combine a climate dataset with an epidemiological model of climate suitability for VBD to obtain suitability projections, to assess the contributions of different sources of climate uncertainty, and to visualize results. See the package documentation for detailed usage examples.

2. Epidemiological model (`epimod`) subpackage: provides classes and methods for models of climate suitability for VBD, in which a suitability metric (e.g., the basic reproduction number, R_0) is defined as a function of temperature and/or precipitation. Methods are also provided for parameterizing mechanistic suitability models by fitting the temperature dependence of model parameters describing vector and pathogen traits to laboratory data (Mordecai et al., 2017). A selection of climate-VBD suitability models from the literature are also available as built-in examples (Kaye et al., 2024; Mordecai et al., 2017; Parham & Michael, 2010; Ryan et al., 2019; Taylor et al., 2019; Villena et al., 2022; Yang et al., 2009).
3. Accessor class for xarray datasets (`xarray.Dataset.climepi`): provides methods for combining climate data with epidemiological models and for assessing and visualizing the importance of different climate uncertainty sources (Hart et al., 2025; Hawkins &

Sutton, 2009), as well as other supporting functions.

4. Front-end application (app) subpackage: provides a method for running the front-end application locally.

Example applications

In a recent study, we used climepi to assess the relative contributions of climate scenario uncertainty and climate model uncertainty, as well as internal climate variability, to uncertainty in future climate suitability for dengue virus transmission in a range of locations that do not currently experience substantial outbreaks (Hart et al., 2025). The climepi documentation includes a detailed usage example demonstrating how results from that study can be reproduced. Further example pages show how climepi can be used to recreate results from two other studies (that did not originally use climepi) (Kaye et al., 2024; Mordecai et al., 2017): specifically, these examples involve parameterizing the temperature-dependent dengue virus transmission suitability model developed by Mordecai et al. (2017) and recreating the analysis of the impact of internal climate variability on climate suitability for *Aedes aegypti* (a vector of VBDs including dengue and yellow fever) by Kaye et al. (2024).

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